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Marine benthopelagic food webs of the North Sea (1997-2015)

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Abstract

A major barrier to assessing the effects of global change on complex ecosystems is the scarcity of food web observations spanning the spatial and temporal scales needed to reveal significant change. This study hopes to rectify this by presenting 4728 empirically derived food webs constructed using a 50 km spatial resolution across the North Sea annually from 1997-2015. We also present a dynamic, reproducible framework for devising food webs at different spatial or temporal scales by integrating nearly four decades of standardised groundfish survey data with one of the world’s most comprehensive stomach content databases. This time series of spatially resolved food webs should help enrich analyses of environmental change on food web structure and aid reliable benchmarking of ecosystem-based management in one of the world’s most commercially important and anthropogenically influenced continental shelves.

Background

Food webs are ecological networks of species interactions that can provide perspective on the structure and functioning of ecosystems¹⁻⁴. Food webs contain nodes which are organisms and edges that represent the flow of energy between organisms due to feeding interactions. Food webs have been used for many decades as a means to understand how environmental stressors alter ecological communities, providing insight into ecosystem complexity and resilience in the face of global change^{1,5-8}. Food webs are also attractive to policy makers and ecosystem managers as they can be used to condense complex ecosystem dynamics into simple metrics, enabling clear advice and the establishment of management benchmarks⁹. For example, The Convention for the Protection of the Marine Environment of the North-East Atlantic¹⁰ uses food webs as a means to assess how human activities are affecting environmental status^{11,12,13,14}. This is particularly important because management has historically often focused on single species or direct pressures, which may overlook indirect effects that propagate through trophic

interactions. By capturing these wider ecological dependencies, food webs provide a means to assess whole-ecosystem health and can therefore support ecosystem-based management.

Given the increase of multiple anthropogenic pressures on marine environments^{14,15}, tools to analyse system-level characteristics such as resilience are paramount to avoid unsustainable impacts and ecosystem degradation. The northeast Atlantic is subject to widespread anthropogenic pressures and yet remains home to extremely productive fisheries^{16,17}. As a result, the North Sea has seen a widespread shifts in biodiversity^{18–21}, ultimately impacting trophic interactions^{22–24}, making it a great case study for the impacts of global change on food webs. However, the extensive data needed to construct food webs, e.g. on interactions between different species across their various life stages and geographic ranges, are often not available which is likely to have been a major limiting factor in food web research. Assessing food webs empirically, especially in open marine ecosystems, is often unachievable due to the unfeasible level of sampling required to effectively characterise trophic interactions. The data that are available are often fragmentary, producing localised and static representations of space and time that can obscure the variability of trophic interactions. Consequently, a common critique of food web research is its lack of robust baselines and thus comparability to novel environmental contexts^{25,26}. Recent methodological advances however, in sampling trophic interactions as well as collaborative efforts to publish large scale data has provided an opportunity to address such challenges. In this study, we make use of such recent advances and further address challenges within food web research by (a) generating an unprecedented number of food webs over multiple decades across the North Sea, and (b) proposing a dynamic, reproducible methodology for food web construction that can be applied consistently across different spatial and temporal scales. We focused on the North Sea to capture a substantial time-series that has not yet been possible at larger scales. Other datasets, such as FISHGLOB²⁷ and additional fish stomach-content compilations²⁸ could support extending food-web construction beyond the spatial scale considered here.

This dataset includes 4728 well-sampled empirically derived food webs using a 50 km spatial resolution across the North Sea over 25 years (Figure 1). Unlike traditional grid-based approaches, these food webs capture overlapping taxa and interactions by sampling within spatial and temporal radii. Constructed using novel big data and a reproducible, flexible methodology, our approach resolves incomplete predator diets, integrates both directly observed and literature-inferred predator-prey interactions, and ensures that trophic structures reflect locally observed communities. The resulting dataset captures northeast Atlantic ecosystem variability at a resolution not previously available,

offering robust baselines for benchmarking and facilitating cross-scale comparisons of trophic interactions. For example, it could be used to ask how human modification of marine ecosystems might mediate climate-driven responses in food web structure and function. The hope is that this resource will fill a key research gap by addressing the need for empirically derived food webs at relevant spatial and temporal scales, support advances in food web research and ecosystem-based management in a region heavily impacted by human activity, while also offering a transferable methodology for application in other marine systems where similar observations exist.

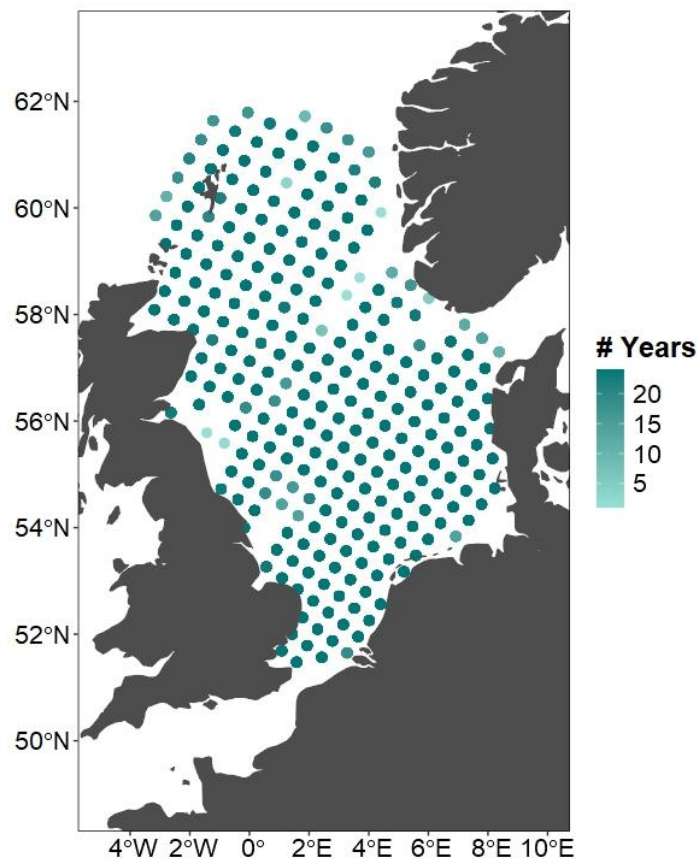


Figure 1: A map of the location of all 4728 food webs in the northeast Atlantic. Each point is the location of a food web, and replicates of each food web exist yearly (where possible) from 1997 to 2015. The colour of the point indicates the number of yearly replicates in each location.

Methods

Description of datasets used to construct the food webs

Fish survey

This dataset offers a standardised record of groundfish catches across the Northeast Atlantic spanning nearly four decades²⁸, drawing on extensive scientific trawl survey data from the ICES database of trawl surveys (DATRAS)²⁹ (DOI: 10.14466/CefasDataHub.126). A subset of -3.5 to 8.5 degrees longitude and 51.24 to 61.75 degrees latitude from 1997-2015 was used in this study due to the more limited availability of standardised data on commercial fishing as well as the extensive time-series available within this region. Here, the subsetted data are referred to as the fish survey database, which holds 3,334,387 abundance observations of 371 taxa sampled from 24,843 hauls from 1997-2015. The most common species in the fish survey are whiting (*Merlangius merlangus*), common dab (*Limanda limanda*), and haddock (*Melanogrammus aeglefinus*). The dataset reports taxa abundance at regular body size intervals and has been supplemented with information on biomass via routine sampling of length-mass relationships. This dataset was used to establish the fish predator taxa present in the environment for any given food web. Unfortunately, marine mammals and seabird data was not included due to limited availability at similar spatial and temporal scales.

Stomach contents

To establish the trophic interactions between consumers and resources, we used a data collation (DOI: 10.14466/CefasDataHub.102)³⁰ that includes DAPTSOM³¹ (DOI: 10.14466/CefasDataHub.144) and ICES Year of the Stomach^{32,33} (DOI: 10.17895/ices.pub.4626). This database is one of the most comprehensive archives of dietary data globally. Stomach samples were collected as early as 1837 with new data continuing to be added. The dataset used here brings together over 415,658 records of trophic interactions within 156,452 individual stomachs sampled from 6924 research hauls across the northeast Atlantic. The most common stomach samples are from Atlantic cod (*Gadus morhua*), whiting (*Merlangius merlangus*) and haddock (*Melanogrammus aeglefinus*). More information on this stomach contents data is detailed by Thompson et al.³⁴. From here on, this dataset is referred to as the stomach contents database.

Steps were taken to clean and standardise the database. Firstly, predator and prey taxa were renamed if there were obvious discrepancies, for example “Octopus” was renamed to “Octopodidae” to match the Latin nomenclature used for the vast majority of observations. Unrefined data whereby the taxonomic resolution was not resolved to the family level or lower, including prey taxa labelled as “Decapoda”, “Chordata”, “Mollusca”, and “Arthropoda” were removed from the entire stomach contents database (13.37 % of original dataset was deleted). Feeding guilds group predators based on similar feeding traits (size and abundance of prey in a diet) and were added to all fish predators in the stomach contents data

to account for ontogenetic shifts in a species diet³⁴. These feeding guilds consisted of planktivore (PL), benthivore (BE), benthopiscivore (BP), and piscivore (PI). If the feeding guild could not be established directly, it was inferred by a predator of the same genus that was the most similar in size. However, predators that had no feeding guild information and less than ten observations across the entire stomach contents were removed from all datasets (0.05 % of original dataset was deleted), as there was not enough information to directly infer their diets.

Food web construction

Food webs of the North Sea were constructed by: (1) defining the temporal and spatial resolution of potential food webs, in this case yearly 50 x 50 km grid cells, (2) constructing species master lists with all species potentially present for each food web, (3) identifying local predator diets, assessing diet completion, and inferring missing trophic links, (4) inferring prey diets, and (5) estimating energy fluxes. In summary, the North Sea was divided into 50 × 50 km grid cells, with each cell centroid used as the focal coordinate. For each focal coordinate and year, fish species observed in trawl surveys within a 100 km radius were identified. The species master list was generated by combining these fish species with taxa recorded in the stomach contents database within the same spatial radius. No temporal subsetting was applied to the stomach contents records because, although the database spans many years, observations are relatively sparse through time. Local predator–prey interactions were then identified from stomach contents data, diet completeness was assessed using species accumulation curves, and missing trophic links were inferred iteratively from taxonomically similar predators where required. Prey diets were inferred using stomach contents data for fish prey and published North Sea food-web information for non-fish prey, before final food webs were assembled with node attributes, including abundance and biomass. Finally, trophic link weights were estimated using energy flux calculations to produce the final food-web database. These steps are summarised in Figure 2, and a more detailed explanation of each step follows below.

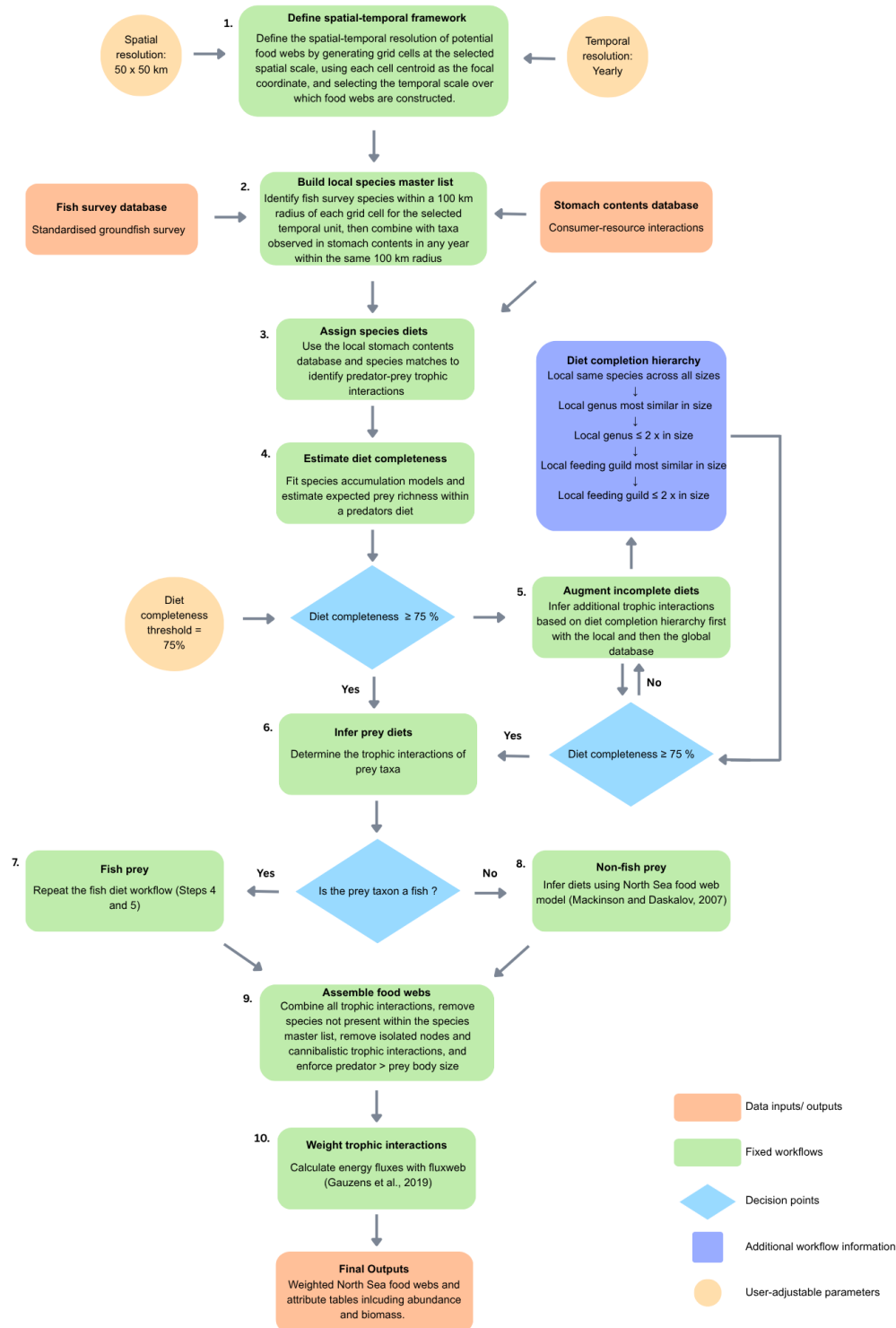


Figure 2: A flow chart summarising the data needed, steps and decisions taken as well as user-adjustable parameters used to construct the North Sea food webs.

1. Define the spatial and temporal resolution of potential food webs

The North Sea study area (-3.5° to 8.5° longitude and 51.25° to 61.75° latitude) was divided into 50×50 km grid cells. The central longitude and latitude of each cell were used as focal points for constructing food webs. An annual temporal resolution was selected to construct food webs with year-specific information from fish survey data.

2. Construct the species master list

Fish species included in each food web were identified from all fish survey hauls conducted within a 100 km radius of the focal coordinate for a given year (1997–2015). This approach captured spatial-temporal variation in the species composition of fish and generated fuzzy boundaries between food webs, with overlapping 100 km radii introducing spatial continuity and dynamic connectivity that are not represented by traditional hard-boundary delineations. To account for differences in sampling effort across the spatial region, a minimum of five and a maximum of ten hauls were taken per 100 km radius. If there were more than ten hauls present, the closest ten to the focal grid coordinate were selected. The majority of grid cells were represented by the maximum number of hauls, with 88% containing ten hauls, while 2% contained the minimum of five hauls. The fish species present were assigned a feeding guild following Thompson et al.³⁴. The mean biomass (kg km^{-2}), abundance (individuals km^{-2}), and mean weight (g) within the 100 km radius were also recorded for each species. The next step involved subsetting the stomach contents database by the same 100 km radius around each focal grid coordinate to identify all possible prey taxa present in that particular space. Here, using a spatial-temporal resolution matching the fish survey data was not possible because the stomach contents database was collected more sporadically through time and did not cover the full study period, particularly the most recent years. Therefore, only spatial variation in stomach contents information was incorporated in the food webs. This, together with the subsetting fish survey data, determined the possible nodes within each food web and is referred to as the species master list i.e. the union of taxa from the fish survey and prey taxa from the stomach-content database within each spatial-temporal unit. Food webs were only constructed for focal grid-cell coordinates and years with fish survey data available within a 100 km radius, and for which stomach-content records were available within the same radius in at least one year.

3. Identify local predator diets, assess diet completion, and infer missing trophic links

Predator–prey interactions were first determined by subsetting the local stomach contents data according to each predator species identified in the fish survey. To ensure sufficient stomach content data were available to accurately characterize predator diets, when establishing trophic interactions, data from all years were combined for each local area. Consequentially when food webs are constructed, they rely on spatial variation of trophic interactions. This approach was necessary because the temporal coverage of stomach content observations was too patchy to reliably assess temporal changes in trophic interactions, with temporal information available for only 7% of the central coordinates. This approach could be easily modified in our code if a different temporal resolution of trophic interactions is desired or becomes available. Most predators in the stomach contents database are resolved to species level, making it straightforward to connect species in the food web based on this information. Where taxa were resolved only to genus level, they were reassigned to a species within the same genus that was also present in the dataset and most similar in size, in order to consolidate nodes and trophic links within the food web. If no other taxa of the same species were present, then the taxa remained resolved to the genus level.

To evaluate whether sufficient data were available to represent each predator's diet, species accumulation (yield–effort) curves were constructed using the 'specaccum' function from the *vegan* R package³⁵ (Figure 3). The expected total prey diversity for each predator was then estimated by fitting six model distributions ('lomolino', 'arrhenius', 'gleason', 'gitay', 'asympt', and 'logis') using the 'fitspecaccum' function, with the best-fitting model selected based on the lowest Akaike Information Criterion (AIC)³⁶ value. A predator's diet was considered to be at a satisfactory level of completion when the observed number of prey taxa represented at least 75% of the total prey richness predicted by the selected model. If the predator species was not represented in the local stomach contents or its diet completion was below 75 %, then local interactions from predator species within the same genus that were most similar in size were included in the diet of that predator. This process was repeated iteratively until all predator diets were sufficiently complete: first by adding local data from all species of the same genus that were not more than double in size; then from local data from species in the same feeding guild and most similar in size; then from local data and all species within the same feeding guild that were not more than double the size; finally from across the global stomach contents database by adding interactions observed for that predator species from the entire range of space and time. The 75% diet-completion threshold was selected as a pragmatic balance between representing the majority of a

predator's expected diet and retaining local dietary specificity. Lower thresholds risked accepting poorly characterised diets, whereas higher thresholds increased reliance on broader genus-, feeding-guild-, or global-level diet substitutions, which could homogenise predator diets and reduce spatial specificity. Therefore, 75% was used to ensure high diet completeness while retaining local trophic structure. Statistics on how many diets were completed at each step are included in Table 1. On average across all food webs, 89 % of fish predator species had a diet completion over 75 %.

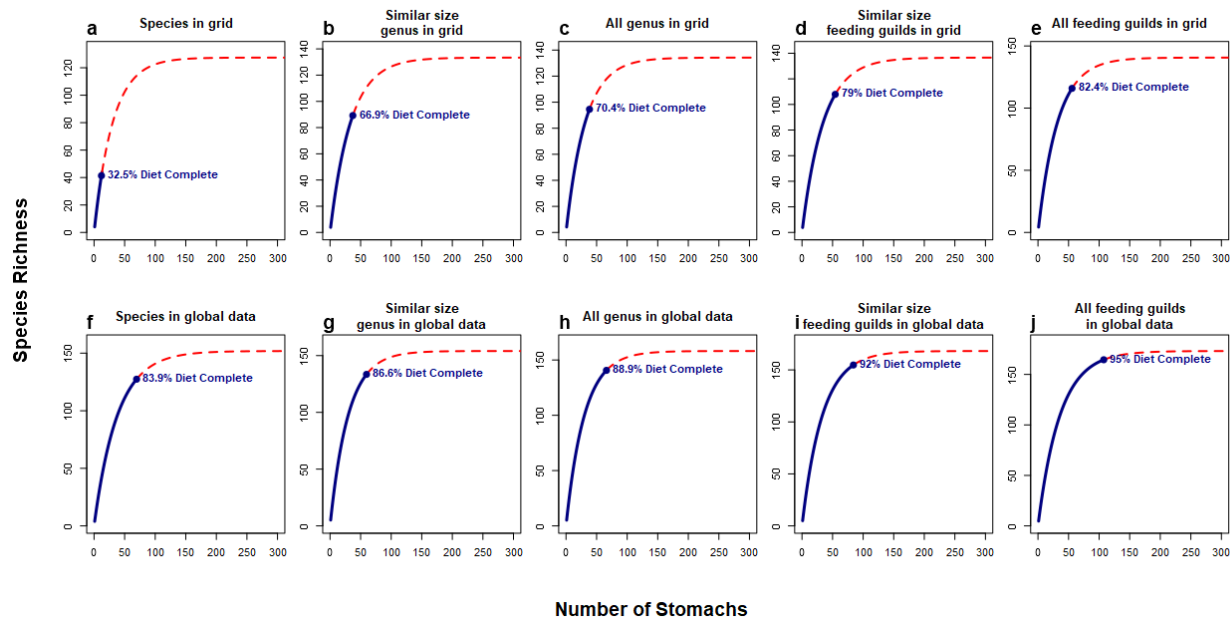


Figure 3: An example of species accumulation curves at each stage of food web construction for *Sprattus sprattus* at longitude -1.1 and latitude 59.3. The blue solid line indicates the observed prey species richness in the stomach contents and the dashed red line is the predicted prey species richness. A satisfactory level of diet completion was achieved when the empirical richness was 75% or more of the predicted prey species richness, i.e. in panel (d), thereby negating the need to add any further links to the diet from panels (e-j).

Table 1: The mean percentage of predators that reached a diet completion of above 75% across all food webs within the study, as determined by the species accumulation curve models.

Level of dietary resolution	Percentage complete	Cumulative percentage complete
Within the same year	0	0
Species within 100 km radius	0	0
Genus most similar in size within 100 km radius	16.6	16.6
Genus not more than double in size within 100 km radius	1.8	18.4
Feeding guild most similar in size within 100 km radius	12.2	30.6
Feeding guild not more than double in size within 100 km radius	4.6	35.2
Species within global stomach contents database	32.1	67.4
Genus most similar in size within global stomach contents database	5.9	73.2
Genus not more than double in size within global stomach contents database	0.4	73.6
Feeding guild most similar in size within global stomach contents database	5.5	79.1
Feeding guild not more than double in size within global stomach contents database	9.8	89.0

The body mass and abundance of the fish predators were taken directly from the local fish survey. The abundance of the non-fish prey was taken from the total prey count across predator stomachs within each food web. The biomass of each prey taxon was then calculated using this abundance multiplied by the individual prey weight. Individual prey weight refers to the average wet weight of a single prey item. The mean individual prey weight for each prey taxon was then calculated across predator

stomachs and multiplied by the total prey count across predator stomachs to estimate total prey biomass within each food web. Biomass estimates therefore represent prey biomass aggregated across predator stomachs within each food web. These measurements have some limitations such as the impact of digestion on estimates of prey weight; however, they are the best estimate in the absence of biological survey data for prey taxa at the same spatial-temporal scale as the fish.

4. Infer prey diets

After establishing trophic interactions of the fish predator species, we attempted to determine trophic interactions between their prey. First, if the prey taxa were themselves fish, the previous step 2 of identifying interactions from the stomach contents database at the species, genus, and feeding guild level were completed for these taxa and their trophic adjacency matrices were constructed. Prey taxa that were not fish were assigned to one of the following feeding groups: herbivorous zooplankton, omnivorous zooplankton, decapods, hydrozoans, tunicates, barnacles, bivalves, bryozoans, omnivorous echinoderms, herbivorous echinoderms, herbivorous gastropods, omnivorous gastropods, cephalopods, omnivorous cnidarians, herbivorous cnidarians, and sponges and their diets were inferred based on a published North Sea food web model by Mackinson and Daskalov^{37,38}. The inferred diet of each grouping is summarised in Table 2. For example, if a prey taxon was classified as omnivorous zooplankton taxon, it was assumed to feed on herbivorous zooplankton taxa based on the North Sea model^{37,38}. Accordingly, a feeding interaction was established between the omnivorous zooplankton taxa and all herbivorous zooplankton known to occur within the spatial scale of the food web. If the prey taxon was known to consume small teleosts^{37,38}, feeding interactions were created to all fish species that were under 10 cm in length at maturity. Some prey taxa consumed basal resources including phytoplankton, coarse particulate organic material (CPOM), fine particulate organic matter (FPOM), microalgae, macroalgae fungi, and bacteria^{37,38}, which are ubiquitous throughout the North Sea and thus were added to every food web.

Table 2: The different prey taxa groupings and their inferred diet from the North Sea Model by Mackinson and Daskalov ^{37,38}.

Prey Group	Diet
Herbivorous zooplankton	FPOM, Bacteria, Fungi, Phytoplankton
Omnivorous zooplankton	Herbivorous zooplankton, Omnivorous zooplankton, FPOM, Bacteria, Fungi, Phytoplankton
Decapods	Herbivorous zooplankton, Omnivorous zooplankton, CPOM, Decapoda, Bivalves, Herbivorous echinoderms, Omnivorous echinoderms, Hydrozoa, Herbivorous gastropoda, Omnivorous gastropoda
Hydrozoans	Herbivorous zooplankton, Omnivorous zooplankton
Tunicates	FPOM, Bacteria, Fungi, Phytoplankton
Barnacles	FPOM, Bacteria, Fungi, Phytoplankton
Bivalves	FPOM, Bacteria, Fungi, Phytoplankton
Bryozoans	FPOM, Bacteria, Fungi, Phytoplankton
Omnivorous echinoderms	Herbivorous zooplankton, Omnivorous zooplankton, Barnacles, Fungi, Macroalgae, CPOM, Bivalves
herbivorous echinoderms	FPOM, Bacteria, Fungi, Phytoplankton
Herbivorous gastropods	FPOM, Bacteria, Fungi, Phytoplankton
Omnivorous gastropods	Herbivorous gastropoda, Omnivorous gastropoda, Bivalves, Bryozoa, Hydrozoa, Tunicate, Fungi, Macroalgae, CPOM
Cephalopods	Small teleosts, Herbivorous echinoderms, Omnivorous echinoderms, bivalves, decapoda, Herbivorous gastropoda and Omnivorous gastropoda
Omnivorous cnidarians	Herbivorous zooplankton, Omnivorous zooplankton, Small teleosts
Herbivorous cnidarians	FPOM, Bacteria, Fungi, Phytoplankton
Sponges	FPOM, Bacteria, Fungi, Phytoplankton

All trophic interactions of the fish predators, fish prey and non-fish prey taxa were combined to complete the food webs. Additional final checks included the removal of any isolated nodes and

cannibalistic trophic links. Cannibalistic links were considered any trophic links coming from and going to the same node. These links were identified as the values in the diagonal of the adjacency matrix and deleted during the food web construction workflow. A final rule whereby no consumer could be smaller than their prey was implemented. This was due to the highly size-structured nature of aquatic ecosystems^{39,40}, supported by a minimum predator-prey body mass ratio of 1.002 across the entire stomach contents database. A final attribute table was developed including taxon names (nodes), feeding group, abundance (N), and biomass (M) as well as the latitude, longitude, and year of the food web in question. The final feeding groups included fish, basal, and all the prey groups listed in Table 2. This created a total of 4865 empirically derived food webs across the North Sea from 1997 – 2015 (Figure 1). The largest food web included 320 nodes and 20,724 trophic links, and the smallest food web included 50 nodes and 543 trophic links.

5. Estimate energy flux

A final step was to calculate the weight of the trophic interactions. This was done using the fluxing function in the *fluxweb* package in R⁴¹. The predictions produced by *fluxweb* are based on body mass scaling of trophic interactions^{41,42}. The model assumes a steady state system where losses by predation or physiological processes equal gains from consumption (1).

$$\sum_j W_{ji} F_i e_{ij} = X_i + \sum_j W_{ij} F_j.$$

equation (1)

where W_{ij} defines the proportion of F_j that is obtained from species i , F_i is the sum of all ingoing energy flux from species i , e defines a species feeding efficiency and X_i is the energetic loss from species i . The weight of each trophic link was added to the list of trophic interactions. Note that energy flux could not be calculated for 137 food webs due to the model being unable to balance the energy flux with the given food web structure, thus the final number of food webs in our database is 4728. An example of a food web can be seen in Figure 4.

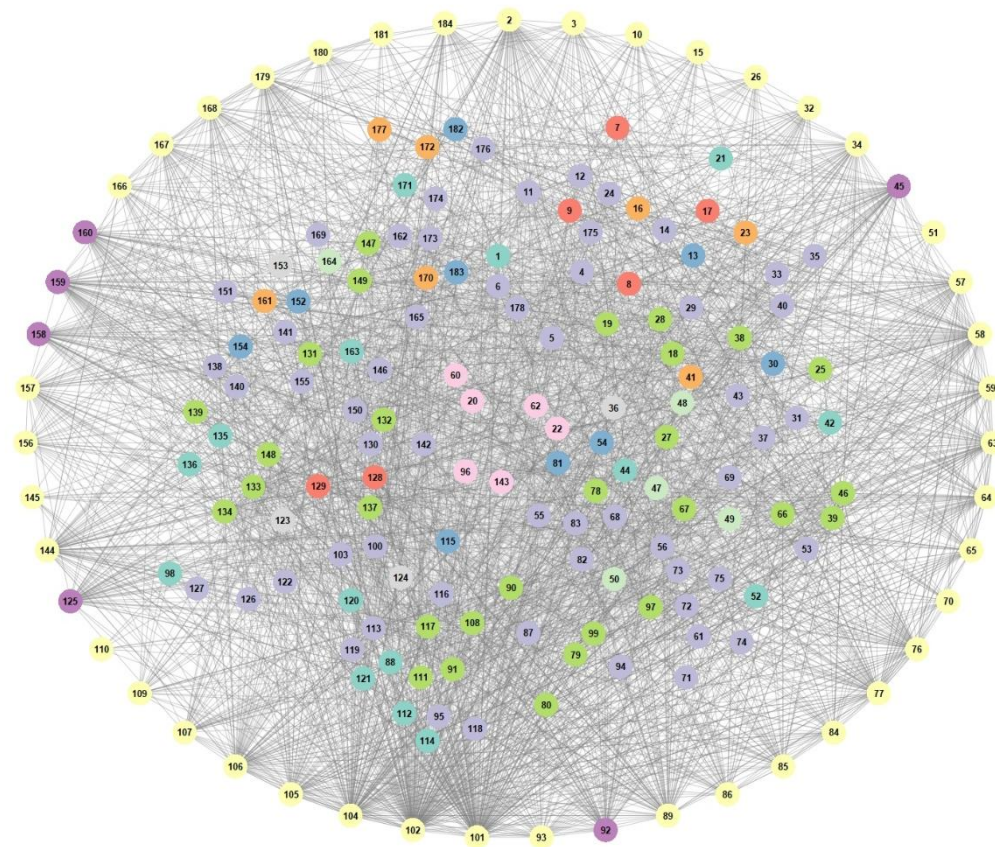
[illegible]

Figure 4: An example food web constructed within a 100 km radius of -1.37 degrees longitude and 59.84 degrees latitude in 2005. The food web was visualised using an adaptation of the ‘PlotWagonWheel’ function in the *cheddar* package⁴³ that allowed all basal resources to be plotted at the centre of the network. Each node is a taxon and the edges drawn between nodes are trophic interactions. If the species belonged to a feeding guild, then it is denoted by an underscore followed by either PI (Piscivore), PL (Planktivore), BP (Benthopiscivore) or BE (Benthivore). The nodes are labelled by numbers and coloured by feeding group, with definitions provided in the graphical legend. The links are weighted by the amount of energy flux (g year^{-1}) calculated using the ‘fluxing’ function in the *fluxweb* R package³³.

Data Record

The North Sea food webs can be found on the University of Essex Data Repository⁴⁴ (<https://dx.doi.org/10.5526/ERDR-00000242>). The food webs are saved as two r.data files: one named FWTL.rdata, which contains a list of all 4728 trophic adjacency matrices; and the other named FWFIn.rdata, which includes a list of all 4728 attribute tables. The attribute tables include columns called nodes (taxon names), M (body mass), N (abundance), feeding group, year, longitude, and latitude (the central coordinates of the focal grid).

Technical Validation

A key validation step involved the use of yield-effort (species accumulation) curves to assess the completeness of each fish predator’s diet. This ensured that the sampling completion of each fish predator’s diet was at least 75% of what could be expected (Figure 1). At each step of the food web

construction, we calculated the percentage completion of each predator in each food web, with the mean results presented in Table 1. In preliminary work using smaller than 100 km radii to sample a predator's diet, completion statistics were noticeably lower hence we chose 100 km as a balance between available data and ecological realism. We also attempted to minimise any differences due to sampling bias or changes in sampling effort across the region by taking a minimum of five and a maximum of ten hauls from the fish survey per food web. Furthermore, the food webs were not based on predetermined sampling location but were placed in a standardised way by gridding the study region and finding coordinate focal points. Food webs were also visually inspected using the *cheddar* package in R⁴³ to observe any inconsistencies or outliers.

Usage notes

The code for reconstructing food webs across spatial and temporal scales is available on the authors' GitHub (https://github.com/amyshurety/Northeast_Atlantic_Food_Web_Construction), and the corresponding data required are found in the University of Essex Data Repository (<https://dx.doi.org/10.5526/ERDR-00000242>). Due to the dynamic nature of the 100 km radius used to collate surveys conducted near each focal grid coordinate, adjacent food webs share data so spatial and temporal autocorrelation should be expected. In this food web compilation, we constructed food webs that were 50 km apart but were representations of a 100 km radius each year. However, care was taken to produce a methodological approach that could be readily applied to new data and/or spatial and temporal scales. The space between, size of the radius, spatial and temporal scale of the food webs can be altered by changing the variables indicated in Table 3. If monthly subsetting of the fish survey is required, here we only subset by year, then lines 122-145 of the R code should be included into the workflow.

Efforts were made to standardise sampling effort across the fish survey dataset; however, when using this dataset, potential bias arising from variable sampling effort in the stomach-contents database should be considered. Future development of this dataset will aim to standardise sampling effort across all contributing datasets.

We hope that this dynamic methodology and the database of North Sea food webs pioneers new research into how ecosystem structure and function vary across large temporal-spatial scales, and in response to different environmental drivers, thereby underpinning robust ecosystem-based management of heavily stressed ecosystems.

Table 3: Variables use to construct different spatial and temporal elements of each food web.

Variable name	Description
grid_m	Number of kilometres used to divide the North Sea into grid squares (here we used 50,000 m i.e. 50 km)
grid_radius	Number of kilometres used as a radius from the centre of the focal grid coordinate (here we used 100,000 m i.e.100 km)
year_min	Minimum year of required fish survey which ranges from 1997 to 2020 (here we used 1997)
year_max	Maximum year of required fish survey which ranges from 1997 to 2020 (here we used 2015)
lat_min	Minimum latitude of required fish survey which ranges from 36.0013 to 61.3535 degrees (here we used 51.25 degrees)
lat_max	Maximum latitude of required fish survey which ranges from 36.0013 to 61.3535 degrees (here we used 61.75 degrees)
long_min	Minimum longitude of required fish survey which ranges from -15.2333 to12.6190 degrees (here we used -3.5 degrees)
long_max	Maximum longitude of required fish survey which ranges from -15.2333 to12.6190 degrees (here we used 8.5 degrees)
year_min_stom	Minimum year of required stomach contents data which ranges from 1836 to 2016 (here we used -Inf to include all the data)
year_max_stom	Maximum year of required stomach contents data which ranges from 1836 to 2016 (here we used -Inf to include all the data)
lat_min_stom	Minimum latitude of required stomach contents data which ranges from 43.6405 to 80.0830 degrees (here we used -Inf to include all the data)
lat_max_stom	Maximum latitude of required stomach contents data which ranges from 43.6405 to 80.0830 degrees (here we used Inf to include all the data)
long_min_stom	Minimum longitude of required stomach contents data which ranges from -56.6000 to 49.9665 degrees (here we used -Inf to include all the data)

long_max_stom Maximum longitude of required stomach contents data which ranges from -56.6000 to 49.9665 degrees (here we used Inf to include all the data)

Code availability

The code used to construct the food webs is available here (https://github.com/amyshurety/Northeast_Atlantic_Food_Web_Construction). Users have a series of parameters, such as the temporal and spatial resolution to generate food webs, that they can adjust, as described in Table 3.

References

1. Pimm, S. L. The complexity and stability of ecosystems. *Nature* **307**, (1984).
2. Kortsch, S. *et al.* Food-web structure varies along environmental gradients in a high-latitude marine ecosystem. *Ecography* **42**, 295–308 (2019).
3. Scharler, U. M. *et al.* Resilience Measures in Ecosystems and Socioeconomic Networks. in *Systems Analysis Approach for Complex Global Challenges* (eds Mensah, P., Katerere, D., Hachigonta, S. & Roodt, A.) 183–208 (Springer International Publishing, Cham, 2018). doi:10.1007/978-3-319-71486-8_11.
4. O’Gorman, E. J. *et al.* A simple model predicts how warming simplifies wild food webs. *Nat. Clim. Change* **9**, 611–616 (2019).
5. Elton, C. S. *Animal Ecology*. (Sidgwick and Jackson, London, UK, 1927).
6. Lindeman, R. L. Seasonal Food-Cycle Dynamics in a Senescent Lake. *Am. Midl. Nat.* **26**, 636–673 (1941).
7. Odum, W. E. & Heald, E. J. The detritus-based food web of a mangrove estuarine community. 265–285 (1975).
8. Link, J. S. What does ecosystem-based fisheries management mean. **27**, 18–21 (2002).

9. Korpinen, S. *et al.* Food web assessments in the Baltic Sea: Models bridging the gap between indicators and policy needs. *Ambio* **51**, 1687–1697 (2022).
10. OSPAR, 2023. *Food Webs Thematic Assessment*. In: *OSPAR, 2023: Quality Status Report*. <https://oap.ospar.org/en/ospar-assessments/quality-status-reports/qsr2023/thematic-assessments/food-webs-ta/> (2023).
11. Schückel, U. *et al.* *Pilot Assessment of Ecological Network Analysis Indices*. <https://oap.ospar.org/en/ospar-assessments/quality-status-reports/qsr-2023/indicator-assessments/pilotassessment-ecological-network-analysis-indices/> (2022).
12. Thompson, M. S. A. *et al.* A feeding guild indicator to assess environmental change impacts on marine ecosystem structure and functioning. *J. Appl. Ecol.* **57**, 1769–1781 (2020).
13. Lynam, C. P., Piet, G. & Volwater, J. Size Composition in Fish Communities. <https://oap.ospar.org/en/versions/2365-en-1-0-0-size-composition-fish-communities/> (2022).
14. Díaz, S. M. *et al.* *The Global Assessment Report on Biodiversity and Ecosystem Services: Summary for Policy Makers*. (Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services, 2019).
15. Jouffray, J.-B., Blasiak, R., Norström, A. V., Österblom, H. & Nyström, M. The Blue Acceleration: The Trajectory of Human Expansion into the Ocean. *One Earth* **2**, 43–54 (2020).
16. Halpern, B. S. *et al.* A Global Map of Human Impact on Marine Ecosystems. *Science* **319**, 948–952 (2008).
17. Halpern, B. S. *et al.* Spatial and temporal changes in cumulative human impacts on the world's ocean. *Nat. Commun.* **6**, 7615 (2015).
18. Hahn, S. J., Brandt, A. & Sonnewald, M. Climate change and fisheries affect benthic composition and diversity in the North Sea — investigations at the Dogger Bank during three decades (1991–

2021) | Marine Biodiversity | Springer Nature Link.

<https://link.springer.com/article/10.1007/s12526-025-01539-8> (2023).

19. Holland, M. M. *et al.* Major declines in NE Atlantic plankton contrast with more stable populations in the rapidly warming North Sea. *Sci. Total Environ.* **898**, 165505 (2023).
20. Püts, M., Kempf, A., Möllmann, C. & Taylor, M. Trade-offs between fisheries, offshore wind farms and marine protected areas in the southern North Sea – Winners, losers and effective spatial management. *Mar. Policy* **152**, 105574 (2023).
21. Gordó-Vilaseca, C., Stephenson, F., Coll, M., Lavin, C. & Costello, M. J. Three decades of increasing fish biodiversity across the northeast Atlantic and the Arctic Ocean. *Proc. Natl. Acad. Sci.* **120**, e2120869120 (2023).
22. Duffy, J. E. *et al.* The functional role of biodiversity in ecosystems: incorporating trophic complexity. *Ecol. Lett.* **10**, 522–538 (2007).
23. Hunt, G. L., Polunin, N. V. C., Engelhard, G. H., Wigham, B. D. & Pinnegar, J. K. Century-Scale Changes in the Feeding Patterns of Demersal Fish Species in the Western North Sea. *Fish Fish.* **27**, 99–115 (2026).
24. Ludwig, K. E., Singer, A., Kröncke, I. & Sell, A. F. Predator-prey trait associations and feeding preferences of demersal fishes in the southern North Sea. *Mar. Ecol. Prog. Ser.* **739**, 173–190 (2024).
25. Niquil, N. *et al.* Reaction of an estuarine food web to disturbance: Lindeman’s perspective. *Mar. Ecol. Prog. Ser.* **512**, 141–154 (2014).
26. Le Guen, C. *et al.* Assessing the ecological status of an estuarine ecosystem: linking biodiversity and food-web indicators. *Estuar. Coast. Shelf Sci.* **228**, 106339 (2019).
27. Maureaud, A. A. *et al.* FISHGLOB_data: an integrated dataset of fish biodiversity sampled with scientific bottom-trawl surveys. *Sci. Data* **11**, 24 (2024).

28. Lynam, C. & Riberio, J. A data product derived from Northeast Atlantic groundfish data from scientific trawl surveys 1983-2020. Cefas
<https://doi.org/https://doi.org/10.14466/CefasDataHub.126> (2022).
29. ICES. ICES Database on Trawl Surveys (DATRAS). <https://datras.ices.dk/>.
30. Thompson, M. S. A., Pontalier, H., Spence, M., Pinnegar, J. & Greenstreet, S. Trophic interaction data and taxonomic information for fish predators and their prey spanning the North East Atlantic and its marginal seas between 1836-2013. Centre for Environment, Fisheries and Aquaculture Science <https://doi.org/10.14466/CEFASDATAHUB.102> (2020).
31. Pinnegar, J., Cooper, K. & Thompson, M. S. A. DAPSTOM (integrated database and portal for fish stomach records). Preprint at <https://doi.org/10.14466/CEFASDATAHUB.144> (2023).
32. Daan, N. *Data Base Report of the Stomach Sampling Project 1981*. 150 (1981).
33. ICES. *Data Base Report of the Stomach Sampling Project 1981*. https://ices-library.figshare.com/articles/report/Data_base_report_of_the_stomach_sampling_project_1981/18624686/1 (1989) doi:10.17895/ices.pub.4533.
34. Thompson, M. S. A. *et al.* Fish functional groups of the North Atlantic and Arctic oceans. *Earth Syst. Sci. Data* **17**, 2447–2462 (2025).
35. Oksanen, J. *et al.* The vegan package. Community ecology package. (2007).
36. Akaike, H. A new look at the statistical model identification. *IEEE Trans. Autom. Control* **19**, 716–723 (1974).
37. Mackinson, S. & Daskalov, G. An ecosystem model of the North Sea to support an ecosystem approach to fisheries management: description and parameterisation. **142**, (2007).
38. Daskalov, G. & Mackinson, S. TROPHIC MODELLING OF THE NORTH SEA. (2007).
39. Woodward, G. *et al.* Body size in ecological networks. *Trends Ecol. Evol.* **20**, 402–409 (2005).

40. Petchey, O. L., Beckerman, A. P., Riede, J. O. & Warren, P. H. Size, foraging, and food web structure. *Proc. Natl. Acad. Sci.* **105**, 4191–4196 (2008).
41. Gauzens, B. *et al.* fluxweb: An R package to easily estimate energy fluxes in food webs. *Methods Ecol. Evol.* **10**, 270–279 (2019).
42. Vucic-Pestic, O., Rall, B. C., Kalinkat, G. & Brose, U. Allometric functional response model: body masses constrain interaction strengths. *J. Anim. Ecol.* **79**, 249–256 (2010).
43. Hudson, L. N. *et al.* Cheddar: analysis and visualisation of ecological communities in R. **4**, 99–104 (2013).
44. Shurety, A. L. *et al.* Marine Benthopelagic Food Webs of the North Sea. University of Essex <https://doi.org/10.5526/ERDR-00000242> (2026).

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Author contributions

A. L. S., E. J. O., M. S. A. T., and E. C. conceptualised and developed the research idea. J. K. P constructed and provided the stomach contents database. M. S. A. T., and E. C., provided the fish survey database. A. L. S., designed and performed the food web construction, which were reviewed by E. J. O., M. S. A. T., E. C., J. K. P and T.C. The writing of the publication was led by A. L. S., and E. J. O. All authors prepared, reviewed and edited the final drafts.

Competing Interests

The authors declare no competing interests.